<table>
<tr><th colspan="4" rowspan="2">Question</th><th rowspan="2">Marking details</th><th colspan="6">Marks available</th></tr>
<tr><th>AO1</th><th>AO2</th><th>AO3</th><th>Total</th><th>Maths</th><th>Prac</th></tr>
<tr><td>6</td><td>(a)</td><td></td><td></td><td>Is a [light] {string / thread} with a {bob / mass / weight} attached to one end</td><td>1</td><td></td><td></td><td>1</td><td></td><td>1</td></tr>
<tr><td></td><td>(b)</td><td></td><td></td><td>Acceleration is proportional to its {distance / displacement} (1)
and directed towards its {equilibrium / fixed} point (1)</td><td>2</td><td></td><td></td><td>2</td><td></td><td></td></tr>
<tr><td></td><td>(c)</td><td></td><td></td><td>[System A] the angle is too big [for shm to occur] (1)

Any × (1) from:
the shorter length {has a larger [percentage] uncertainty / is less accurate}
the shorter period {has a larger [percentage] uncertainty / is less accurate}</td><td></td><td></td><td>2</td><td>2</td><td></td><td>2</td></tr>
<tr><td></td><td>(d)</td><td>(i)</td><td></td><td><table>
<tr><th>l /m</th><th>$\sqrt{l}$ / $\mathrm{m}^{\frac{1}{2}}$</th><th>T_{20} /s</th><th>Mean T_{20} /s</th><th>Uncertainty T_{20} /s</th></tr>
<tr><td>0.250</td><td>0.50</td><td>19, 21, 22, 20</td><td>21</td><td>2</td></tr>
<tr><td>0.500</td><td>0.71</td><td>26, 29, 26. 29</td><td>28</td><td>2</td></tr>
<tr><td>0.750</td><td>0.87</td><td>35, 34, 33, 36</td><td>35</td><td>2</td></tr>
<tr><td>1.000</td><td>1.00</td><td>42, 39, 40, 41</td><td>41</td><td>2</td></tr>
<tr><td>1.250</td><td>1.12</td><td>45, 42, 44, 45</td><td>44</td><td>2</td></tr>
<tr><td>1.500</td><td>1.22</td><td>48, 50, 49, 52</td><td>50</td><td>2</td></tr>
</table>
[1 for each column]
Deduct a maximum of 1 mark overall for inconsistent sig figs in each column</td><td></td><td>3</td><td></td><td>3</td><td>3</td><td>3</td></tr>
</table>

Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
					Marks available					
		(ii)		See example graphs at end of Q6 Award 3 marks for all points and error bars correct Award 2 marks for 5 points and error bars correct Award 1 mark for 4 points and error bars correct Award 0 marks for 3 points or less and error bars correct Correct lines drawn for the maximum gradient and the minimum gradient (1) **ecf**		4		4	4	4
		(iii)		l can be measured to ± 0.001 [m] or resolution is 1 mm (1) [Uncertainty in $\sqrt{l}$] is too small to plot (1)			2	2		2
		(iv)		Both gradients correct (1) Mean gradient correct (1) **ecf** Absolute uncertainty in gradient correct (1) **ecf** implied by % uncertainty Correct g (1) **ecf** Final % correct (1) **ecf No sig fig or unit penalties** $m_{\text{max}} \left[= \frac{58.2}{1.40 - 0.06} \right] = 43.4$; $m_{\text{min}} \left[= \frac{54.3 - 4.7}{1.4} \right] = 35.4$ $\therefore$ Gradient, m = 39.4 ± {4.0 / 10%} **Either** For 20 oscillations, $T = 20 \times 2\pi\sqrt{\frac{l}{g}}$ $\therefore \sqrt{g} = \frac{40\pi}{m}$ [or by impl.] [= 3.19 ± 10%] $\therefore$ g = 10.2 m s^{-2} ± 20%		5		5	5	5

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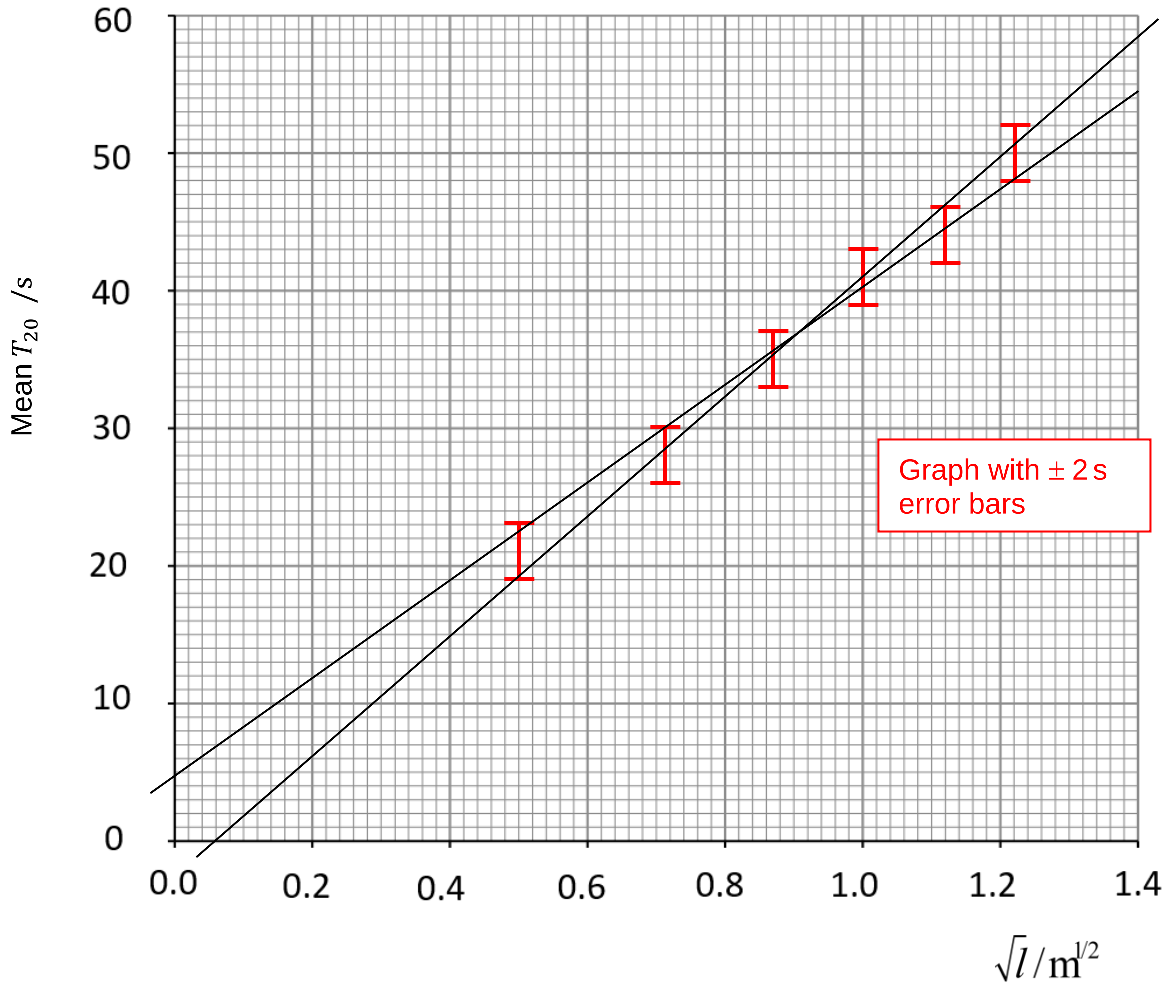
<tr><td></td><td></td><td></td><td></td><td>Or
For one oscillation: m = 1.97 ± 10% (1)
$\left[T = 2\pi\sqrt{\frac{l}{g}}\right]$, $\therefore\ \sqrt{g} = \frac{2\pi}{m}$ [or by impl.] [= 3.19 ± 10 %] (1)
$\therefore$ g = 10.2 (± 20 %) m s^{-2}

Alternative: calculate max value of g from min gradient; min value of g from max gradient and hence g ± uncertainty.

Gradients of both: 43.284 and 35.357 (1)
Mean $T_{20} = \frac{40\pi}{\sqrt{g}}\sqrt{l}$ so gradient = $\frac{40\pi}{\sqrt{g}}$
$g = \left(\frac{40\pi}{43.3}\right)^2 = 8.42\ \mathrm{m\ s^{-2}}$ and
$g = \left(\frac{40\pi}{35.3}\right)^2 = 12.67\ \mathrm{m\ s^{-2}}$ (1)
So $\mathrm{unc}(g) = \frac{12.67-8.42}{2} = 2.1\ \mathrm{m\ s^{-2}}$ (1)
$g = 10.5\ \mathrm{m\ s^{-2}}[1]$
$\therefore$ % unc(g) = $\left[= \frac{2.1}{10.5} \times 100\%\right]$ = 20% (1)</td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
</table>

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(e)			Slope $\propto \frac{1}{\sqrt{g}}$ **or** equivalent e.g. gradient = $\frac{2\pi}{\sqrt{g}}$ **or** gradient = $\frac{40\pi}{\sqrt{g}}$ (1) Ratio of slopes = $\frac{1}{\sqrt{0.2}}$ **or** $\sqrt{5}$ **or** new gradient = 89 or two correct expressions for gradients e.g. $\frac{2\pi}{\sqrt{g}}$ **and** $\frac{2\pi}{\sqrt{0.2g}}$ **or** $\frac{40\pi}{\sqrt{g}}$ and $\frac{40\pi}{\sqrt{0.2g}}$ (1) Ratio of gradients = 2.24ish and sensible comment (1)			3	3	3	
				Question 6 total	**3**	**12**	**7**	**22**	**15**	**17**


60
50
40
30
20
10
0
Mean T_{20} /s
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
$\sqrt{l}$ / m$^{1/2}$
Graph with ± 2 s error bars